Selective sparse coding and borrow-tolerant masks for a smaller ten-integer consistency equation

Mathematical research prepared for ProveIt

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1 A selective, borrow-tolerant ten-integer equation

There is a completely fixed polynomial in ten unrestricted integer unknowns

$$F_{10}(f, g, h, k, l, w, x, y, z, m)$$

of exact total degree 21 111 400, with a division-free evaluation circuit using 659 binary operations, for which ordinary mathematics formalizable in ZFC proves

$$\neg\exists (f, g, h, k, l, w, x, y, z, m) \in \mathbb{Z}^{10}\ (F_{10} = 0) \quad \Longleftrightarrow \quad \mathrm{Con(ZFC)}.$$

The proof uses the pinned ProveIt machine/compiler calibration and Sun's arithmetic bridge. It is not an end-to-end proof-assistant certificate and it does not establish either consistency or inconsistency. The unknown count is inherited, not a new universal unknown-count theorem.

Three source-layer changes are responsible for the improvement: selective isolation of used quadratic monomials, one-bit target padding that tolerates incoming borrows, and calibration by the actual mask population instead of a uniform worst-case carry threshold. The latter two changes are proved below and are independent of any search heuristic. All scalar powers in this section have fixed nonnegative integer exponents. Auxiliary letters are polynomial abbreviations, not extra unknowns.

1.1 Fixed data and selective extraction

Use precisely the 22 quadratic residuals of `data/parent_quartic31.json` and the complete fixed port tuple

$$(\beta_{\mathrm{native}}, K_0, M_C, M_F) = (\texttt{beta}, \texttt{K}, \texttt{MC}, \texttt{MF_source}).$$

The last port is already shifted; it must not be shifted again. Its tuple is Wang943 121×8801, with port bit lengths $(1064921, 1070004, 1064921, 1064922)$. Each payload is included, with compressed and uncompressed hashes. The active binding is `INSTANCE.json`; historical relative paths inside preserved input files are provenance only.

Replace every positive parent coordinate by $n_i + 1$. Let $r_j(n)$ denote the resulting residual, written $r_j = \sum_\alpha p_{j,\alpha} n^\alpha$, $|\alpha| \le 2$. This is the original package-31 source, not the later package-45 transport variant. The inherited source theorem is

$$\exists n \in \mathbb{N}^{31} \bigwedge_{j=1}^{22} r_j(n) = 0 \quad \Longleftrightarrow \quad \neg\,\mathrm{Con(ZFC)}, \qquad \mathbb{N} = \{0, 1, 2, \ldots\}.$$

Its parent coordinate order is

$$(J, F, \alpha, z, f, h, i, o, j, s, w, \tau, \eta, \zeta, y, \gamma, \\ N, E, P, c, a, X, D, C_2, L, F_2, V, g, I, B_{\mathrm{aux}}, \Delta).$$

Assign these coordinates the following weights, with a padding weight $a_0 = 0$:

$$(a_0, \dots, a_{31}) = (0, 9, 25, 23, 139, 35, 151, 168, 21, 91, 181, 94, 52, 150, 180, 7,\\ 135, 38, 88, 183, 106, 178, 136, 20, 138, 8, 64, 105, 154, 37, 137, 120).$$

The residual order and corresponding targets are

$$(16, 2, 9, 14, 21, 7, 15, 22, 13, 3, 19, 20, 1, 10, 17, 8, 4, 6, 11, 18, 12, 5),\\ (70, 86, 451, 685, 1050, 1296, 1530, 1874, 2212, 2358, 2696, 2780,\\ 2792, 3157, 3290, 3634, 3929, 4238, 4384, 4527, 4865, 5190).$$

Write q_j for the target assigned to residual r_j, and put

$$t = 183, \quad q = 5190, \quad K_c = q + 2t = 5556, \quad E_c = t + K_c + 2 = 5741, \quad H_c = 2t + K_c + 3 + 22 = 5947.$$

For every used α and every γ with $|\gamma| \le 2$, the finite certificate verifies

$$q_j - \alpha \cdot a + \gamma \cdot a = q_k \quad \Longleftrightarrow \quad j = k \text{ and } \alpha = \gamma. \tag{1}$$

All reversed exponents are nonnegative. There are 136 used source terms, 528 possible padded quadratic monomials, and $136 \cdot 528 = 71808$ checked interactions. Only 267 different pair sums occur: a full Sidon condition is deliberately not required.

Define

$$c_{j,\alpha} = (2 - |\alpha|)! \prod_i \alpha_i!\, p_{j,\alpha}, \qquad D_c(Z) = \sum_{j,\alpha} c_{j,\alpha} Z^{q_j - \alpha \cdot a}.$$

Condition (1) implies that the 136 displayed exponents in D_c are distinct. Indeed, if two were equal, multiplying by the monomial belonging to either source term would violate isolation at its target. Thus, for $A_c \ge \max |c_{j,\alpha}|$, every coefficient of D_c has absolute value at most A_c, with no multiplicity factor. Its leading coefficient is exactly 2: the last residual is r_5 and $r_5(0) = 1$. In particular, the zero vector is not a source solution.

For $C_n(Z) = (1 + \sum_i n_i Z^{a_i})^2$, multinomial expansion gives

$$[Z^{q_j}] C_n(Z) D_c(Z) = 2 r_j(n), \qquad |[Z^s] C_n D_c| \le A_c (1 + \textstyle\sum_i n_i)^2. \tag{2}$$

The factorial multiplier cancels the multinomial coefficient; isolation excludes every unwanted contribution. This proves the identity for all source vectors, not just tested examples.

Take A_c to be the maximum coefficient majorant obtained by replacing in each fixed-port recipe a monomial by $2^{\sum e_i \ell_i}$, multiplying by the absolute value of its integer coefficient, and summing. Here ℓ_i are the four verified port bit lengths. Exact integer arithmetic gives $\mathrm{bitlength}(A_c) = 2134927$ and, with

$$\kappa = 2134940, \qquad \beta_S = 2^\kappa, \qquad \lfloor \beta_S / A_c \rfloor = 16383,$$

we have

$$\beta_S > 4096 A_c \quad \text{and} \quad \beta_S > 64 A_c. \tag{3}$$

1.2 The new source polynomials

In the two unrestricted integer unknowns f, g, define

$$b = 1 + 3f, \qquad B_c = \beta_S b^2, \qquad \mathcal{C} = b B_c^t,$$

$$\ell = \sum_{i=1}^{31} B_c^{a_i}, \qquad N_0 = B_c^{t+1}, \qquad N_1 = B_c^{K_c+1}, \qquad N = N_0 N_1,$$

$$A_T = \sum_{j=1}^{22} B_c^{q_j}, \qquad M_0 = N_0 - 1 - (b-1)\ell,$$
$$T_c = M_0 + (B_c - 2)A_T N_0, \qquad J_c = 2(1+g)^2 D_c(B_c) + A_T, \qquad S_c = g + J_c N_0,$$
$$X = N^2 + (S_c + T_c)N + N - 1 - S_c,$$
$$Y = \beta_S^{15}\, 2^{\kappa-21}\, B_c^{5931} b.$$

Equivalently, $Y = 2^{\kappa H_c - 21} b^{11863}$; the first form avoids materializing its larger fixed coefficient. The fixed leaf $2^{\kappa-21} = 2^{2134919}$ is explicitly specified, not an unknown. Also $\mathcal{C} = 2^{\kappa t} b^{367}$ has the monomial form required by the inherited bridge.

Bounds before binary-base recovery. Suppose $f \neq 0$, b is an integer square and $0 \leq g < 2\mathcal{C}$. Since $b \equiv 1 \pmod 3$, these hypotheses imply $b \geq 4$ and $f > 0$. The integer B_c is positive and even, but is not yet assumed to be a power of two. Its size implies $B_c > 2b$ and $B_c > A_c + 1$. In the following estimates write $B = B_c$ temporarily.

The base-B digits of M_0 are $B-1$ except at the 31 designated positions, where they are $B-b$. Hence $0 \leq M_0 < N_0$. The leading term of $D_c(B)$ is $2B^q$, and

$$|D_c(B) - 2B^q| \leq A_c \frac{B^q - 1}{B-1} < B^q, \qquad 0 < D_c(B) < \frac{A_c B^{q+1}}{B-1}.$$

Because g is integral, $1 + g \leq 2bB^t$. Therefore

$$0 < 2(1+g)^2 D_c(B) < \frac{8A_c b^2}{B-1} N_1 < \frac{1}{4} N_1.$$

The last inequality follows from $\beta_S > 64A_c$ and $B > 2$. Also $(B-1)A_T < B^{q+1} \leq N_1/B^2$, since $t \geq 1$. Consequently

$$0 < J_c + (B-2)A_T < \tfrac{1}{2} N_1.$$

We have $g < N_0$, so $g + M_0 < 2N_0$. As $N_1 \geq 4$, this proves

$$0 \leq S_c, \quad 0 \leq T_c, \qquad S_c + T_c < N. \tag{4}$$

In particular $X \geq N^2 > 3b$ and $Y > \max(b, 256)$. Furthermore $B_c \equiv N_0 \equiv N \equiv 1 \pmod 3$, $M_0 \equiv 0$ and $T_c \equiv -22$, giving

$$X \equiv 1 + T_c \equiv 0 \pmod 3.$$

These statements were obtained without assuming that a square b is a power of two. This order matters in the reverse bridge.

1.3 Sparse padding that tolerates borrows

Now assume additionally that b is a power of two. Then B_c and N_0 are powers of two. The condition $g \,\&\, M_0 = 0$ and $0 \leq g < N_0$ is exactly

$$g = \sum_{i=1}^{31} n_i B_c^{a_i}, \qquad 0 \leq n_i < b.$$

The weights are distinct, so the tuple is unique. Indeed, the low mask forbids every bit outside the low $\log_2 b$ bits of these designated base-B_c cells.

For this tuple let $h_s = [Z^s] C_n D_c$ and let ϵ_s be the indicator of the 22 target positions. Equations (2) and (3) give $|h_s| < B_c/4$, and

$$J_c = \sum_{s=0}^{K_c} (2h_s + \epsilon_s) B_c^s.$$

This is a signed-coefficient expansion, not yet a digit expansion. Set $u_0 = 0$ and carry out ordinary Euclidean digit normalization:

$$u_{s+1} = \left\lfloor \frac{2h_s + \epsilon_s + u_s}{B_c} \right\rfloor, \qquad d_s = 2h_s + \epsilon_s + u_s - B_c u_{s+1}.$$

Induction gives $u_s \in \{-1, 0\}$ and $0 \le d_s < B_c$. At a marked position $\epsilon_s = 1$, the binary mask $B_c - 2$ permits exactly the digits 0 and 1. If $h_s = 0$, the unnormalized digit is $1 + u_s \in \{0, 1\}$, so it passes. If $h_s \ge 1$, that digit lies in $[2, B_c)$ and fails. If $h_s \le -1$, it lies strictly between $-B_c + 1$ and 0, so its normalized digit is at least 2 and again fails. Thus each marked test detects precisely $h_s = 0$, regardless of an incoming borrow. Positivity of J_c was already proved, so these are ordinary digits of a nonnegative integer.

Concatenation at the power-of-two boundary N_0 now yields

$$S_c \,\&\, T_c = 0 \quad \Longleftrightarrow \quad g = \sum_i n_i B_c^{a_i},\ 0 \le n_i < b, \quad r_1(n) = \cdots = r_{22}(n) = 0. \tag{5}$$

The dense geometric padding used by the previous construction has been replaced by only one extra bit at each marked cell. Unmarked negative coefficients may generate borrows; the proof allows them rather than silently assuming them absent.

1.4 Calibrating the binomial threshold by the mask population

For $N = 2^u$ and nonnegative integers S, T satisfying $S + T < N$, set

$$X_* = N^2 + (S + T)N + N - 1 - S.$$

Its leading, middle, and low blocks are disjoint. If s_2 denotes binary population, then

$$\begin{aligned} s_2(X_*) &= 1 + s_2(S + T) + u - s_2(S) \\ &= 1 + u + s_2(T) - \big(s_2(S) + s_2(T) - s_2(S + T)\big). \end{aligned}$$

The parenthesized term is nonnegative and vanishes exactly when $S \,\&\, T = 0$. Legendre's formula gives $\nu_2\binom{2X_*}{X_*} = s_2(X_*)$. Consequently

$$2^{1+u+s_2(T)} \mid \binom{2X_*}{X_*} \quad \Longleftrightarrow \quad S \,\&\, T = 0. \tag{6}$$

The stronger range $S + T < N$ is essential and was proved in (4); separate inequalities $S < N, T < N$ would not suffice.

Write $b = 2^k$ and $B_c = 2^v$ on the binary branch. The low mask has population $(t + 1)v - 31k$, while each of the 22 high marked cells contributes $v - 1$ bits. Hence

$$s_2(T_c) = (t + 1 + 22)v - 31k - 22, \qquad u = E_c v,$$

and

$$1 + u + s_2(T_c) = H_c v - 31k - 21 = \kappa H_c + (2H_c - 31)k - 21 = \log_2 Y.$$

Combining (6) and (5) proves the required reverse source implication. Conversely, a source solution is nonzero because $r_5(0) = 1$. Choose arbitrarily large $b = 4^s$ exceeding all its coordinates and put $f = (b - 1)/3$. Its encoding satisfies $b \le B_c \le g < \mathcal{C}$ and all carry tests pass. Thus it supplies arbitrarily large forward witnesses with $Y \mid \binom{2X}{X}$. This establishes the entire source contract for the unchanged finalizer.

1.5 Explicit radical-free finalizer

The following formulas are the inherited package-31 quintic, merged-square, structural-weight finalizer. They are written out to fix the actual equation. Distinguish its local B_*, S_*, T_* from the source base and masks above. Define

$$
\begin{gathered}
L = lY, \qquad U = 2LX, \qquad V = 4gwY, \qquad W = bw, \\
A = U(V+1), \qquad B_* = 2X+1, \qquad C = B_* + (A-2)h, \\
P = U^2V, \qquad K = X + 1 + k(P-2), \\
S_* = 2A-5, \qquad T_* = 3WC - 2(W^2-1), \\
R_2 = (P^2-4)K^2 + 4, \\
\Delta = 8\mathcal{C}^3 g K^2 - g^2\{32(C-KL)^2\mathcal{C}^3 + g^2K^2\}, \\
O = f^2 l^2 \Delta, \qquad \xi = zx, \\
D = (A^2-4)C^2 + 4, \qquad E = C^2 D\xi, \\
F = 4(A^2-4)E^2 + 1, \qquad G_* = 2(A+2)(A-2)^2E^2 - 1, \\
\Omega = 16E, \qquad H_* = C + (B_* - C)F + \Omega F y, \\
I_* = (G_*^2 - 1)H_*^2 + 1.
\end{gathered}
$$

Put

$$
\begin{aligned}
r &= T_* + S_* y, & \Sigma &= \xi S_*^5, & & \\
\Theta &= (r^3 - rS_*^2 - S_*^3)(r^2 + 23S_*^2), & & & & \\
a_0 &= b\{1 + (3O-4)b^2z^2\}, & a_1 &= R_2, & a_2 &= DFI_*, \\
w_0 &= 1, & w_1 &= a_0, & w_2 &= a_0(P^2 + K^2 + 3), \\
Z &= \Sigma m - \Theta, & A_i &= \Sigma^2 w_i^2 a_i \quad (0 \le i \le 2), & & \\
V_n &= Z^2 + A_2 - A_0 - A_1, & W_n &= V_n^2 + 4Z^2A_2 - 4A_0A_1. & &
\end{aligned}
$$

The promised single equation is

$$
\boxed{F_{10} = W_n^2 - 16V_n^2Z^2A_2 = 0.} \tag{7}
$$

Every expression is an integer polynomial. In particular, radicals and binomial coefficients appear in the proof only, never in the equation.

Exact dependency boundary. Sun's 2021 Theorems 4.12–4.13, specialized as in his 2026 Propositions 6.1–6.2, provide the two bridge directions for any source satisfying the contract just proved. Forward witnesses have b square, R_2 square, $S_* \mid T_*$, $C = u_{B_*}(A)$ and $O > 0$. Reverse witnesses with b square, $O > 0$, $\xi \neq 0$, R_2, DFI_* square and $S_* \mid T_*$ recover $b \in 2\uparrow$, $0 < g < 2\mathcal{C}$ and the required binomial divisibility. The residue interpolation theorem permits ξ to be divisible by any prescribed positive integer and permits any residue class for $y \pmod{\xi}$. The only data substituted here are the newly proved source polynomials; the bridge does not require $Y = N^2$.

For completeness, the inherited finalizer implications can be verified as follows. The polynomial $(v^3 - v - 1)(v^2 + 23)$ has a root modulo every positive integer and none over $\mathbb{Q}$. For odd primes other than 23 the cubic discriminant is -23: if the cubic is irreducible its Frobenius permutation is even, so the quadratic has a simple root; otherwise the cubic has a simple root. At 23 the cubic has simple root 3. At 2, $-23 \equiv 1 \pmod 8$. Hensel lifting and the Chinese remainder theorem finish the local-root argument. Prime valuations in the two homogeneous factors show that $\Sigma \mid \Theta$ forces $S_* \mid T_*$: if $v_p(S_*) > v_p(T_*)$, the valuation of Θ is $5v_p(T_*) < 5v_p(S_*)$. Since S_* is odd, it never vanishes; the absence of rational roots ensures $\Theta \neq 0$.

When $z \neq 0$, a_0 square implies that its coprime factors b and $1 + (3O-4)b^2z^2$ are signed squares. The congruence $b \equiv 1 \pmod 3$ excludes a negative square and excludes zero, including the zero-factor exception.

Thus b is a positive square and the usual positivity Pell lemma gives $O > 0$. Conversely, forward witnesses have $O \geq 4$: the positive integer Δ is divisible by $g \geq b \geq 4$. For $d = 3O - 4 > 0$, $d \equiv 2 \pmod 3$ is nonsquare. A positive Pell unit raised to its order modulo b gives $v = bz > 0$ with $u^2 - dv^2 = 1$, making a_0 a square. Residue interpolation then supplies $\xi = zx$ and a y that simultaneously makes a_2 square and $\Sigma \mid \Theta$.

If $z \neq 0$, the displayed a_0 always has $|a_0| > 1$. The exceptional possibilities $a_0 = 0, \pm 1$ force $|b| = 1$, hence $b = 1$, $f = O = 0$, and $a_0 = 1 - 4z^2$, a contradiction. Also $\sqrt{|a_1|} \leq P^2 + K^2 + 2$. Thus the weights obey

$$|w_1| > \sqrt{|a_0|}, \qquad |w_2| > \sqrt{|a_0|} + |w_1|\sqrt{|a_1|}.$$

In the multiquadratic field, an automorphism changing a nonzero weighted root has a largest changed index; its magnitude exceeds the sum of all smaller changed magnitudes. A rational weighted sum can therefore occur only when every a_i is an integer square. This also covers negative radicands by complex absolute values, and zero radicands separately.

The exact norm identity

$$F_{10} = \prod_{\epsilon \in \{\pm 1\}^3} \left(Z + \sum_{i=0}^{2} \epsilon_i \sqrt{A_i} \right)$$

has 35 expanded terms and is checked independently by integer polynomial arithmetic. If $\Sigma = 0$ then $F_{10} = \Theta^8 \neq 0$. At a zero of F_{10} we therefore have $\Sigma \neq 0$, hence $z \neq 0$; separation forces all a_i square and $\Sigma \mid \Theta$. The reverse bridge and source proof recover a parent solution. Conversely the forward construction just described makes all a_i square and $\Sigma \mid \Theta$; choosing m to cancel one signed factor gives $F_{10} = 0$. Composition with the pinned parent theorem proves the stated consistency equivalence in ZFC.

1.6 Exact finite certificates and comparison

The literal DAG `generated/signed10_b2.json` has ten variable leaves, 18 fixed leaves, and 659 live binary arithmetic gates: 214 additions, 26 subtractions and 419 multiplications. Multiplication by a numeral is charged; fixed-coefficient construction is outside this arithmetic-circuit cost. It is not a bit-complexity claim.

The independent checker does not import or execute the builder. It reconstructs the shifted quadratic source and fixed recipes, verifies all selective interactions, checks 52 local polynomial circuit identities exactly, checks the norm identity, and audits liveness and the gate ledger. For the total degree it propagates a syntactic upper bound, then tracks the coefficient of that degree after substituting the same indeterminate for all ten variables. The resulting nonzero residues are

p	101	1009	1000003
$[T^{21111400}]F_{10}(T,\ldots,T) \pmod p$	36	42	374910.

Any one of these gives the exact lower bound, hence the total degree is exactly 21 111 400. This is not a random evaluation argument. Finite small-base audits additionally corroborate the borrow and population lemmas, including non-power-of-two square source bases. They are tests of implementations, not substitutes for the unbounded proofs.

The previous package-40 ten-integer point was degree 237 520 456 with 782 operations. The new degree is lower by a factor exceeding 11.25, and the circuit uses 123 fewer operations, a reduction of about 15.7%. Package 43's 809-operation variant is also improved. These are comparisons of explicit constructions at the pinned repository snapshot, not claims of global optimality, of an absolute "simplest" equation, or of improvement over different variable/domain tradeoffs.

The pinned revision is `124c07c0b77a9e20e84676514d8b1ff78536ea3f`. `SOURCE_PINS.json` records the verified exact source bytes. Primary arithmetic references are Z.-W. Sun, *Further results on Hilbert's Tenth Problem*, Sci. China Math. 64 (2021), 281–306, https://arxiv.org/html/1704.03504v7; and *Ten unknowns for Hilbert's tenth problem over the integers*, 2026 preprint, https://arxiv.org/html/2609.01594v2. The inherited finalizer proof and coefficient adapter are preserved in `upstream/`. Their original relative links describe upstream layouts; the source pin manifest provides their current verified origins.